Complete SQL & PostgreSQL Mastery Guide

A comprehensive guide covering database fundamentals, advanced querying, relationships, and PL/pgSQL procedures for data analytics and backend development.

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1. SQL Basics & CRUD Operations

- **What it is:** SQL (Structured Query Language) is the standard language used to interact with Relational Database Management Systems (RDBMS) like PostgreSQL.

- **Why we use it:** To store, retrieve, update, and manage millions of rows efficiently compared to traditional spreadsheets (Excel).
- **Core CRUD Operations:**
    - **CREATE:** Adding new databases, tables, or records (INSERT INTO).
    - **READ:** Fetching data (SELECT).
    - **UPDATE:** Modifying existing records (UPDATE ... SET).
    - **DELETE:** Removing unwanted records (DELETE FROM).

2. Data Types & Constraints

- **Data Types:** Defining what kind of data a column can hold (e.g., INT for numbers, VARCHAR for text, BOOLEAN for true/false, TIMESTAMP for dates).
- **Constraints (Rules & Validations):**
    - PRIMARY KEY: Uniquely identifies each row in a table (cannot be NULL and must be unique).
    - FOREIGN KEY: Links two tables together to maintain data integrity.

- NOT NULL: Ensures a column cannot have empty values.
- UNIQUE: Ensures all values in a column are different.
- CHECK: Ensures data meets a specific condition (e.g., age >= 18).

3. Clauses, Operators & Aggregations

- **Clauses:** Used to filter and organize result sets.
  - WHERE: Filters rows based on specific conditions.
  - GROUP BY: Groups rows that have the same values into summary rows.
  - ORDER BY: Sorts the result set in ascending (ASC) or descending (DESC) order.
- **Operators:** Comparison (=, >, <), Logical (AND, OR, NOT), and Pattern Matching (LIKE, IN).
- **Aggregation Functions:** Used to perform calculations on multiple rows:
  - COUNT(): Counts total rows.

- SUM(): Calculates the total sum.
- AVG(): Finds the average value.
- MAX() / MIN(): Finds highest and lowest values.

4. String Functions & CASE Statements

- **String Functions:** Manipulating text data within SQL queries (e.g., CONCAT, SUBSTRING, LOWER, UPPER, LENGTH).
- **CASE Statement:** Acts like an if-else condition inside SQL queries to dynamically generate new columns based on specific logic (e.g., categorizing salary ranges into "High", "Medium", "Low").

5. Altering Tables & Schema Management

- **What it is:** Modifying the structure of an existing database table without losing existing data using the ALTER TABLE command.
- **How we use it:** Adding new columns, dropping unnecessary columns, renaming columns, or modifying constraints dynamically as business requirements change.

6. Database Relationships & Joins

- **Types of Relationships:**
  - **One-to-One (1:1):** One record in Table A relates to only one record in Table B (e.g., User and User Profile).
  - **One-to-Many (1:N):** One record in Table A relates to multiple records in Table B (e.g., Customer and their Orders).
  - **Many-to-Many (N:M):** Multiple records in Table A relate to multiple records in Table B, usually connected via a Junction/Mapping table (e.g., Students and Courses).
- **SQL Joins:** Combining rows from two or more tables based on a related column between them:
  - INNER JOIN: Returns records that have matching values in both tables.
  - LEFT JOIN: Returns all records from the left table, and the matched records from the right table.
  - RIGHT JOIN: Returns all records from the right table, and the matched records from the left table.

    - FULL OUTER JOIN: Returns all records when there is a match in either left or right table.

7. Views & Stored Procedures (PL/pgSQL)

- **Views:** A virtual table based on the result-set of an SQL statement. Used for security (hiding sensitive data) and simplifying complex queries for reporting.
- **Stored Procedures (PL/pgSQL):** A saved collection of SQL statements and procedural logic (loops, conditions) stored in the database server. Used for automation, performance optimization, and executing repetitive business logic.